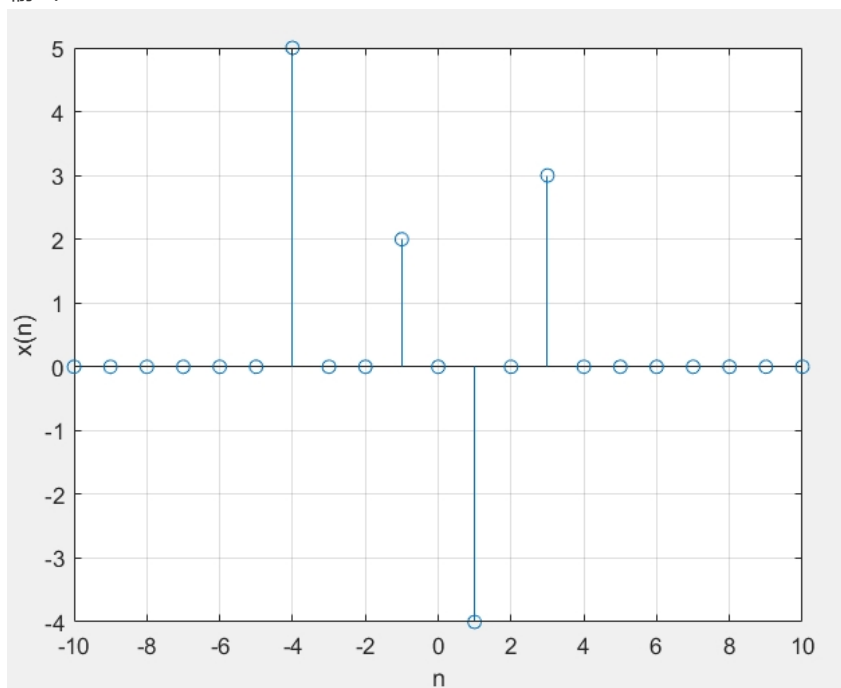


```

n=-10:10;
x=zeros(size(n));
x(n==-4)=5;
x(n==-1)=2;
x(n==1)=-4;
x(n==3)=3;
stem(n,x);
xlabel('n');
ylabel('x(n)');
grid on;

```

输出



```

syms n k alpha beta n0 N;

h(n)=alpha^n*(heaviside(n)-heaviside(N-1));
x(n)=beta^(n-n0)*heaviside(n-n0);

y=symsum(subs(h,n,k)*subs(x,n,n-k),k,-inf,inf);
y=simplify(y);
disp('卷积和 y(n) 为: ');
pretty(y);

```

输出

$$\int_{-\infty}^{\infty} \frac{1}{k} \alpha^k \beta^{n-k-n_0} (\text{heaviside}(N-1) - \text{heaviside}(k)) (\text{heaviside}(k-n+n_0) - 1) dk$$

$$- \beta \frac{n - n_0}{\text{heaviside}(N - 1)} \frac{\beta \text{heaviside}(k + n - n_0)}{\alpha k} \Big|_{k=1}^{\infty}$$

输出

1 / 1


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{
{

#7 <= 0 if ((0 <= #1 and 0 <= #12 n0 and #12 (k - n) <= 0 and 0 <= N)

k
or (#2 n <= 0 and #2 n0 <= 0 and N <= 0 and #2 <= 0 and k == 0))
and 0 <= alpha

#6 <= 0 if ((0 <= #1 and 0 <= #12 n0 and #12 (k - n) <= 0 and 0 <= N)

k
or (#2 n <= 0 and #2 n0 <= 0 and N <= 0 and #2 <= 0 and k == 0))
and alpha <= 0

0 <= #7 if ((0 <= #1 and #12 n0 <= 0 and 0 <= #12 (k - n) and 0 <= N)

k
or (0 <= #2 n and 0 <= #2 n0 and N <= 0 and #2 <= 0 and k == 0))
and 0 <= alpha

0 <= #6 if ((0 <= #1 and #12 n0 <= 0 and 0 <= #12 (k - n) and 0 <= N)

k
or (0 <= #2 n and 0 <= #2 n0 and N <= 0 and #2 <= 0 and k == 0))
and alpha <= 0

0 <= #10 if ((#1 <= 0 and 0 <= #12 n0 and #12 (k - n) <= 0 and 0 <= N)

k
or (#2 n <= 0 and #2 n0 <= 0 and N <= 0 and 0 <= #2 and k == 0))
and 0 <= alpha

0 <= #5 if ((#1 <= 0 and 0 <= #12 n0 and #12 (k - n) <= 0 and 0 <= N)

k
or (#2 n <= 0 and #2 n0 <= 0 and N <= 0 and 0 <= #2 and k == 0))
and alpha <= 0

#10 <= 0 if ((#1 <= 0 and #12 n0 <= 0 and 0 <= #12 (k - n) and 0 <= N)

k
or (0 <= #2 n and 0 <= #2 n0 and N <= 0 and 0 <= #2 and k == 0))
and 0 <= alpha

#5 <= 0 if ((#1 <= 0 and #12 n0 <= 0 and 0 <= #12 (k - n) and 0 <= N)

k

```

[illegible]

$$\begin{aligned}
& \begin{array}{c} | \\ 0 \leq \#3 \text{ if } 0 \leq \#11 \text{ and } \alpha \leq 0 \text{ and } N \leq 0 \text{ and } 0 \leq \#11 (k + n) \text{ and } 0 \leq \\ \#11 n0 \end{array} \\
& \begin{array}{c} | \\ k \\ | \\ \#3 \leq 0 \text{ if } 0 \leq \#11 \text{ and } \alpha \leq 0 \text{ and } N \leq 0 \text{ and } \#11 (k + n) \leq 0 \text{ and } \#11 \\ n0 \leq 0 \end{array} \\
& \begin{array}{c} | \\ k \\ | \\ 0 \leq \#4 \text{ if } \#11 \leq 0 \text{ and } 0 \leq \alpha \text{ and } N \leq 0 \text{ and } 0 \leq \#11 (k + n) \text{ and } 0 \leq \\ \#11 n0 \end{array} \\
& \begin{array}{c} | \\ k \\ | \\ \#4 \leq 0 \text{ if } \#11 \leq 0 \text{ and } 0 \leq \alpha \text{ and } N \leq 0 \text{ and } \#11 (k + n) \leq 0 \text{ and } \#11 \\ n0 \leq 0 \end{array} \\
& \begin{array}{c} | \\ k \\ | \\ 0 \leq \#8 \text{ if } \#11 \leq 0 \text{ and } \alpha \leq 0 \text{ and } N \leq 0 \text{ and } 0 \leq \#11 (k + n) \text{ and } 0 \leq \\ \#11 n0 \end{array} \\
& \begin{array}{c} | \\ k \\ | \\ \#8 \leq 0 \text{ if } \#11 \leq 0 \text{ and } \alpha \leq 0 \text{ and } N \leq 0 \text{ and } \#11 (k + n) \leq 0 \text{ and } \#11 \\ n0 \leq 0 / \end{array}
\end{aligned}$$

where

$$\#1 == \beta \frac{k - n + n0}{k}$$

$$\#2 == \beta \frac{n - n0}{k}$$

$$\#3 == \frac{\alpha^2 \#11 k (k + n)}{k}$$

$$\#4 == \frac{N \#11 k (k + n)}{\alpha k}$$

$$\#5 == \alpha \#12 k^2 (k - n)$$

$$\#6 == \alpha \frac{k^2}{\#12 k n_0}$$

$$\#7 == N \alpha \frac{k}{\#12 k (k - n)}$$

$$\#8 == \frac{\#11 k^2 n_0}{\alpha k}$$

$$\#9 == \frac{N \#11 k n_0}{\alpha k}$$

$$\#10 == N \alpha \frac{k}{\#12 k n_0}$$

$$\#11 == \beta^{k + n - n_0}$$

$$\#12 == \beta^{n - k - n_0}$$